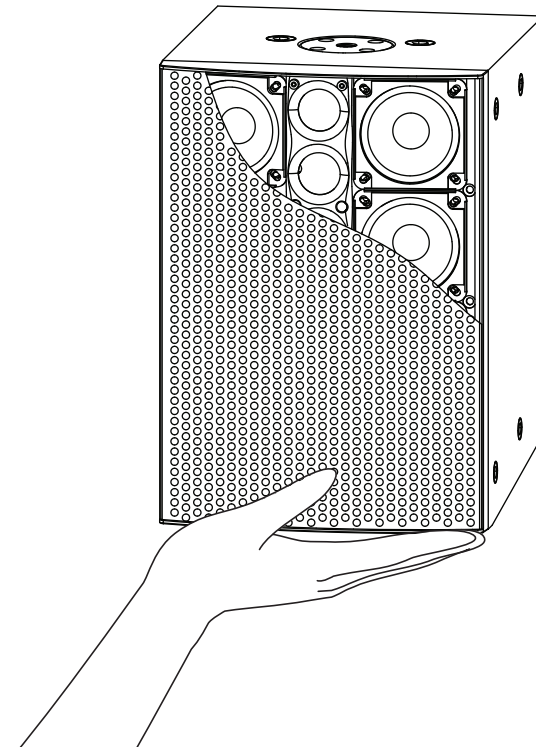




www.se-audiotechnik.com
www.se-audiotechnik.co.kr



USER'S MANUAL



M-F3

SMALL SIZE, HIGH SPL 2.1 ACTIVE **ARRAY SYSTEM**

M-F3

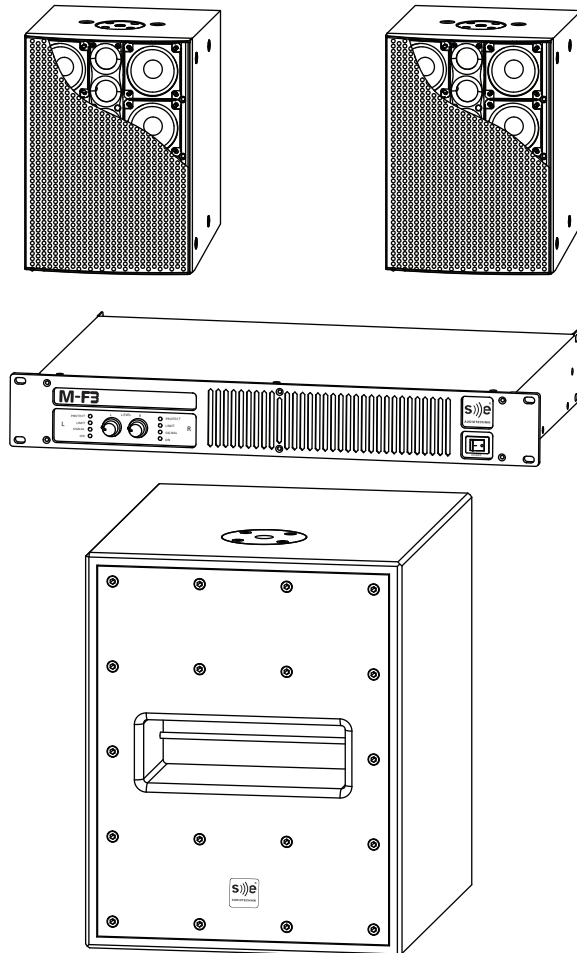
SMALL SIZE, HIGH SPL 2.1 ACTIVE **ARRAY** SYSTEM

Thank you for choosing **s)))e AUDIOTECHNIK** !

We have designed this product to operate reliably over many years. For that **s)))e** guarantees with its name and many years of experience as a producer.

Please, take a few moments to read these instructions carefully, as we want you to enjoy your new **s)))e** products quickly and to the fullest.

You can find more information about **s)))e** audiotechnik at our Internet site **WWW.SE-AUDIOTECHNIK.COM**
WWW.SE-AUDIOTECHNIK.CO.KR



PREVENTIVE MEASURES:

1. Please read the attached safety instructions as well as the following instructions carefully.
2. Please keep all the instructions.
3. Please use the device only as intended.
4. Please respect the valid waste management rules. Please deliver the packaging divided into plastic and paper / cardboard to the recycling management.
5. Please refer all servicing to qualified personnel only if the device is damaged, exposed to liquid / rain or if it does not operate normally.
6. Please do not expose the product to any kind of heat such as ovens, radiators, or any other devices (incl. amplifiers). Please also make sure that there is enough distance between amplifiers and walls, racks, etc. to prevent overheating.
7. After connection please check the wiring to prevent any kind of accident or damage.
Please never use any kind of damaged cable and wiring.
8. Only use approved and stable stands, brackets, shelves, tables etc.. for installations. Please check for adequate stability to avoid a collapse.
9. Appearance of interferences when using wireless systems.
The simultaneous use of wireless microphones and mobile phones (if both devices are not very far away from each other) can cause interferences in the microphone signal which can be heard in the PA system.



CAUTION:

To reduce the risk of electric shock, do not remove cover (or back). No user serviceable parts inside. Refer servicing to qualified personnel.



The lightning flash with arrowhead symbol within an equilateral triangle is intended to alert the user to the presence of uninsulated "dangerous voltage" within the product's enclosure that may be of sufficient magnitude to constitute a risk to persons.



The exclamation mark within an equilateral triangle is intended to alert the user to the presence of important operating and maintenance (servicing) instructions in the literature accompanying the appliance.

CAUTION! HIGH VOLUME!

You will operate this transmission system for professional use. Therefore the commercial use of this equipment is liable to the rules and regulations of the Accident Prevention & Insurance Association of your industry sector. s))e as a manufacturer is bound to inform you formally about the existence of eventual sanitary risks.

This system is able to induce an acoustic pressure of 80 dB. 85 dB is by law the maximum audio pressure level which your ear can be exposed to during a work day. It was set according to the technical expertise of the occupational medicine as a basis for the noise rating level. Higher sound levels or longer exposition times could damage your ear. The time of exposition by higher sound pressure levels should be shortened in order to prevent from ear damages. Here are a few reliable warning signals which show that you have exposed yourself for a too long period to excessive sound pressure levels:

- You hear bell- or whistling sounds!
- You have the impression that you can't hear high tones anymore!

FEATURES & GETTING STARTED:

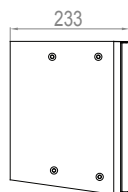
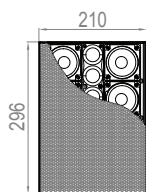
INTRODUCTION

M-F3 is a compact and modular active all-in-one array system, which is composed of one $2 \times 10''$ active sub, one 1.5 U amplifier and two passive full range speakers.

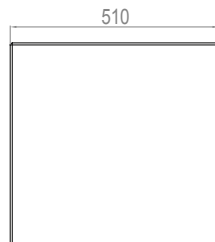
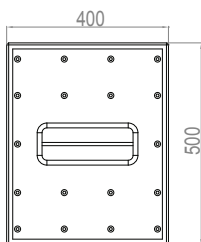
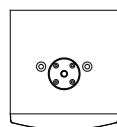
The M-F3 features small size and high power output by adopting line array principles. A combination of the speakers can also be used as a line array system.

With built-in DSP, feedback destroy switch and also high shelf, the amplifier has greatly facilitated the system's ability to conduct tuning and feedback suppression in complex environments.

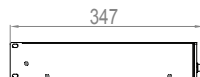
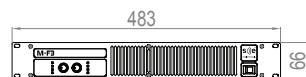
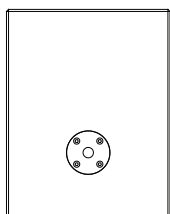
STRUCTURAL DIAGRAM:



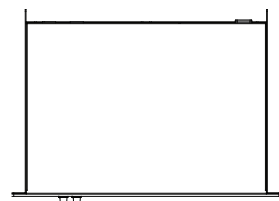
M-F3 SAT



M-F3 SUB



M-F3 AMP



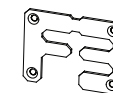
OPTIONAL ACCESSORIES:



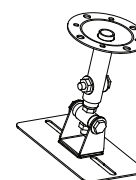
3/8" Adaptor
3/8" Adaptor



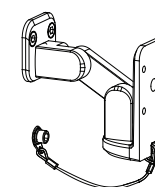
SPS 20
M20 Pole Support



M-F3 LC
M-F3 Link Connector



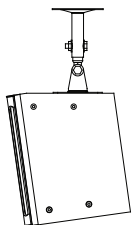
CM
Ceiling Mount



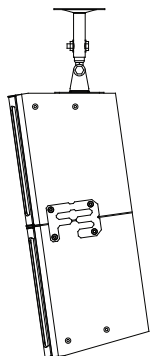
SMB
Smart-Mounting-Bracket

OPERATION OPTIONS:

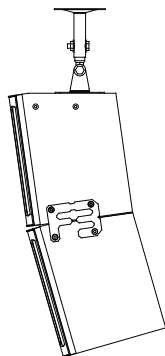
BS 8



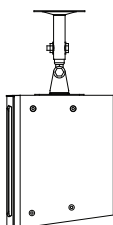
BD 8



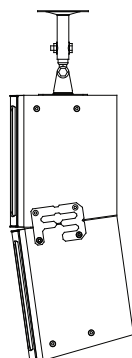
BD 816



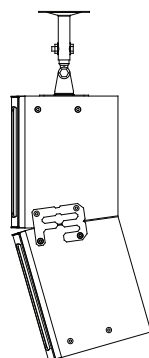
TS 0



TD 08



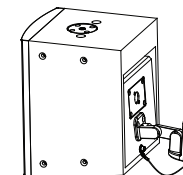
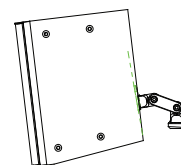
TD 016



1 FLYING

T-Top mounting
B-Bottom mounting
S-Single box
D-Dual boxes
0 / 8 / 16 - Angle

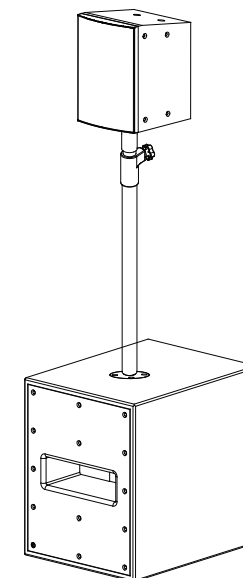
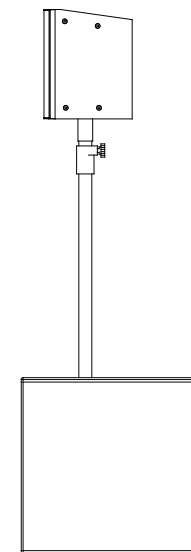
2 WALL MOUNTING



M-F3 SAT 1/pc
SMB 1/pc

3 GROUND STACKING

Ground installation system of
M-F3 SAT and M-F3 SUB with
3 / 8" connecting pole and
SPS 20

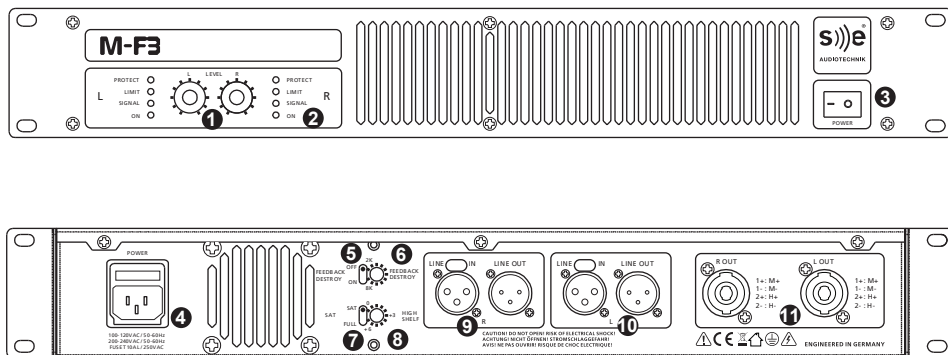


M-F3 SAT 1/pc
M-F3 SUB 1/pc

3/8" Adaptor 1/pc
SPS 20 1/pc

CONNECTIONS, CONTROLS AND INDICATORS:

M-F3 AMP



1 SIGNAL ATTENUATION LEVEL CONTROL KNOBS

Permit independent control of each channel's attenuation.

2 INDICATOR LEDS

ON: Lights up once the system is properly connected to the power mains and switched on.

SIGNAL: Lights up as soon as an audio signal is present. The signal acquisition is performed before the MAIN LEVEL controller.

LIMIT: Lights up when the limiter starts working. If the limiter LED lights up permanently or for longer periods, reduce the volume level. Failure to do so may result in a distorted sound and damage to the speaker system.

PROTECT: Lights up if the amplifier is short-circuited or overheated. The amplifiers are muted automatically. Upon reaching normal operating conditions, the device reverts to normal operating mode after a few seconds.

3 POWER SWITCH

Position - : Connects the amplifier's current feed. (green LED on).

Position 0 : Disconnects the power.

Before switching the active speaker on or off, always adjust the volume to zero (MAIN LEVEL all the way to the left).

4 IEC POWER INLET

An appropriate mains cord is included in the package. The M-F3 system is intended for 2 operating voltage ranges (100 - 120 V AC and 200 - 240 V AC). The conversion is done automatically.

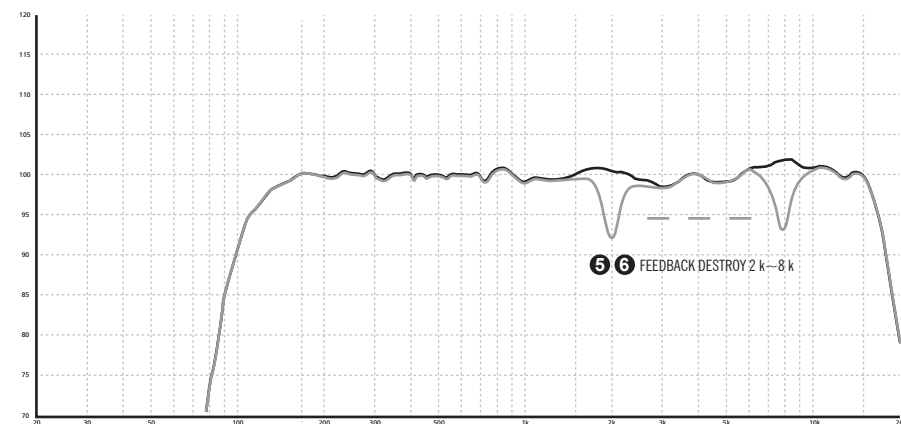
CONNECTIONS, CONTROLS AND INDICATORS:

5 FEEDBACK DESTROY ON / OFF SWITCH

If the feedback destroy switch is in position of ON, it will create a bell shaped dip filter with a width (Q) of 10 and - 8 dB.

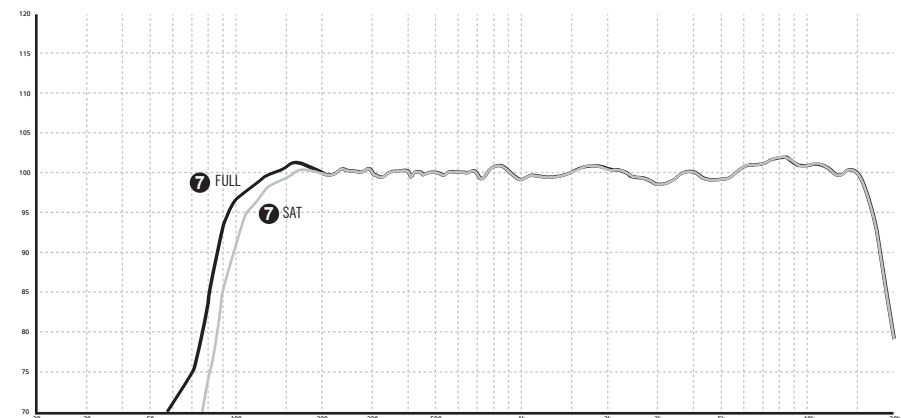
6 FEEDBACK DESTROY FREQUENCY KNOB

Dip frequency center point can be swept from 2 kHz to 8 kHz.



7 SAT / FULL SWITCH

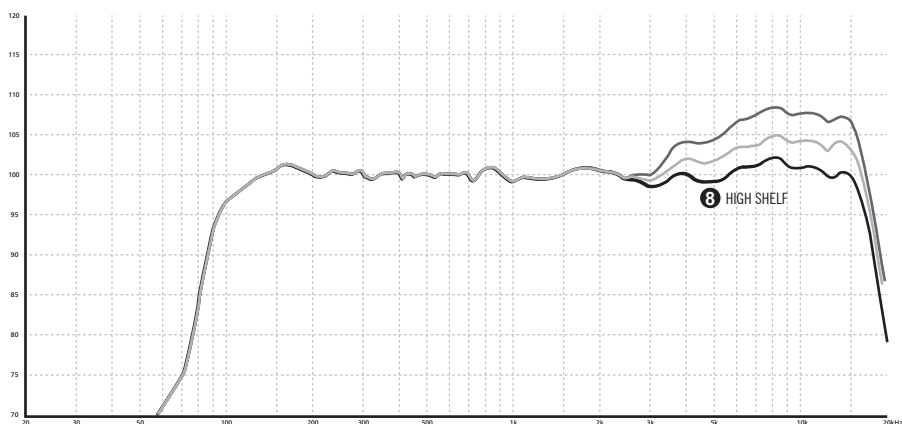
If the SAT / FULL switch is in position of SAT, it will create a high-pass filter which cuts off bass frequency below 120 Hz at a rate of 24 dB / OCT. If the switch is in position of FULL, a high-pass filter at 100 Hz with 24 dB / OCT is used.



CONNECTIONS, CONTROLS AND INDICATORS:

8 HIGH SHELF KNOB

This control gives up to 6 dB boost above 5 kHz.



9 LINE INPUT / OUTPUT RIGHT

Female Neutrik XLR connectors for the amplifier's signal input. Male Neutrik XLR connectors for the other active speakers or amplifiers' signal output. The output signal is driven through a built-in router circuitry, which ensures signal run through the whole path without any loss.

10 LINE INPUT / OUTPUT LEFT

Female Neutrik XLR connectors for the amplifier's signal input. Male Neutrik XLR connectors for the other active speakers or amplifiers' signal output. The output signal is driven through a built-in router circuitry, which ensures signal run through the whole path without any loss.

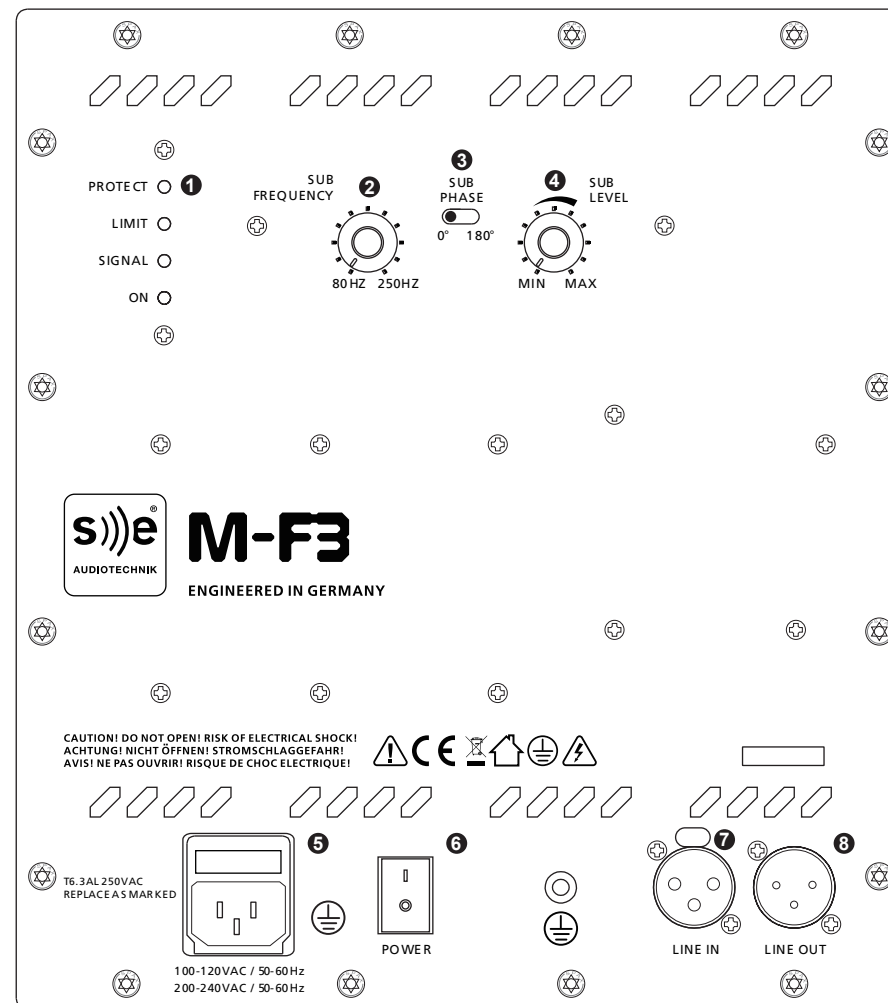
11 SPEAKER OUTPUT CONNECTORS

Neutrik speakon to connect the M-F3 SAT.

- 1+ : M+
- 1- : M-
- 2+ : H+
- 2- : H-

CONNECTIONS, CONTROLS AND INDICATORS:

M-F3 SUB AMP



CONNECTIONS, CONTROLS AND INDICATORS:

1 INDICATOR LEDS

ON: Lights up once the system is properly connected to the power mains and switched on.

SIGNAL: Lights up as soon as an audio signal is present. The signal acquisition is performed before the MAIN LEVEL controller.

LIMIT: Lights up when the limiter starts working. If the limiter LED lights up permanently or for longer periods, reduce the volume level. Failure to do so may result in a distorted sound and damage to the speaker system.

PROTECT: Lights up if the amplifier is short-circuited or overheated. The amplifiers are muted automatically. Upon reaching normal operating conditions, the device reverts to normal operating mode after a few seconds.

2 SUB FREQ

Setting the low pass frequency of the subwoofer between 80 Hz and 250 Hz.

3 SUBPHASE 0° / 180°

The phase reverse switch for the subwoofer (0°, 180°) is factory set to 0°. This setting is suitable for the majority of applications. In unfavourable acoustic conditions, the setting of 180° may have a positive effect on the bass and low mid performance.

4 SUB LEVEL

Volume controller for the M-F3 subwoofer.

5 IEC POWER INLET

An appropriate mains cord is included in the package. The M-F3 system is intended for 2 operating voltage ranges (100 - 120V and 200 - 240 V AC). The conversion is done automatically.

6 POWER SWITCH

Position - : Connects the amplifier's current feed. (green LED on).

Position 0 : Disconnects the power.

Before switching the active speaker on or off, always adjust the volume to zero (MAIN LEVEL all the way to the left).

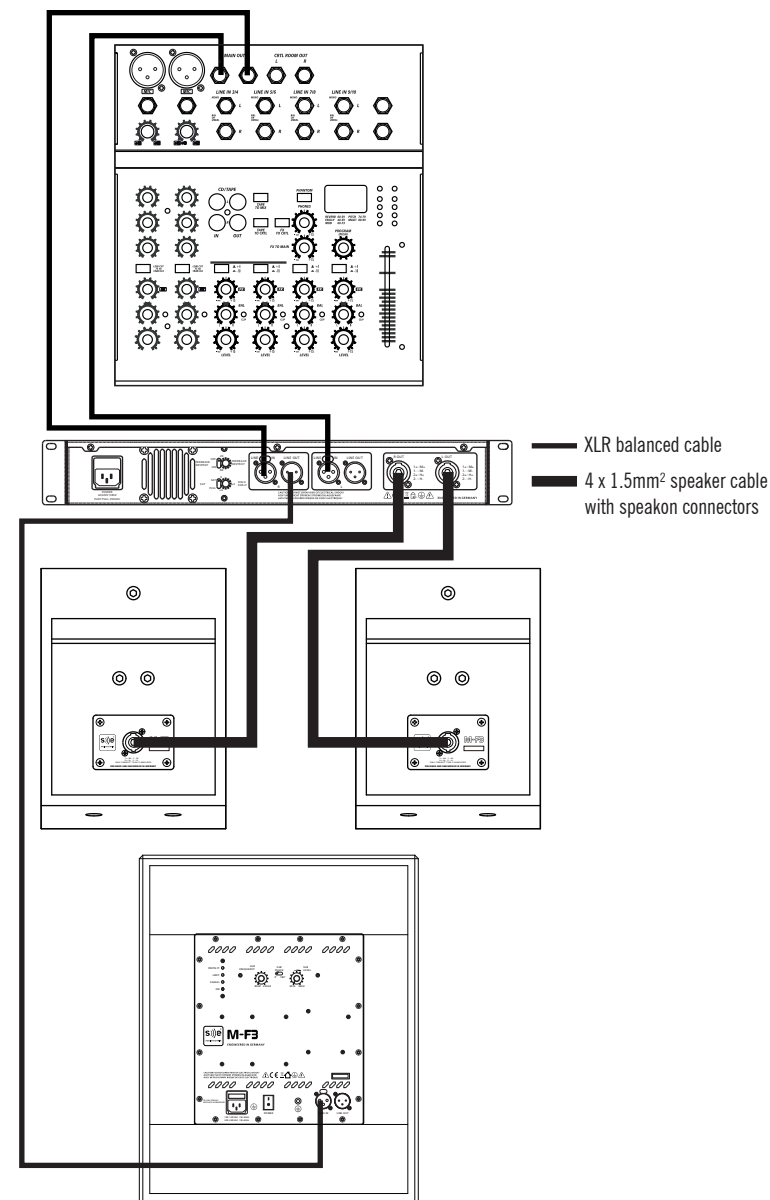
7 LINE INPUT

The balanced line input with female Neutrik XLR connector to connect a playback device (e.g. mixer).

8 LINE OUTPUT

The balanced line output with male Neutrik XLR connector can be used to control other active speakers or amplifiers. The output signal is driven through a built-in router circuitry, which ensures signal run through the whole path without any loss.

WIRING EXAMPLE:



SPECIFICATIONS:

M-F3 SAT	
Type:	2.8" bass reflex
Frequency Response (-6dB):	120 Hz-20 kHz
Sensitivity (1W / 1m):	100 dB
Max. SPL (1m):	128 dB
Dispersion (H × V):	120° × 16°
Transducer MF:	8 × 2.8" Neodymium Magnet with cooling module, 0.75" Voice Coil
Transducer HF:	7 × 1" Neodymium Magnet, 1" Voice Coil
Cabinet / Coating:	12 mm MDF, white PU texture coating
Input:	Neutrik NL- 4
Dimensions (W × H × D):	210 × 296 × 233 mm
Weight:	5.5 kg + 5.5 kg
M-F3 SAT can only be used with the M-F3 AMP.	

M-F3 SUB	
Type:	10" V construction, bandpass
Frequency Response (-6dB):	40 Hz - 250 Hz
Power Output(RMS):	400 W
Max. SPL (1m):	129 dB
Transducer LF:	2 × 10" Ferrite, 2" Voice Coil
Power Amplifier Type:	Class D amplifier, SMPS, 400 W RMS
Signal Processing:	DSP, 48 kHz, 24 bits
Protection:	Multiple limiter, short circuit, overheating
Controls:	On / Off switch, sub level, Low pass frequency knob, 0° - 180° phase switch
Indicators:	On, signal, limit, protect
Line Input / Output:	Input Neutrik XLR / Router-output Neutrik XLR
Cabinet / Coating:	15 mm plywood, black polyurea coating
Dimensions (W × H × D):	400 × 500 × 510 mm
Weight:	25 kg

SPECIFICATIONS:

M-F3 AMP	
Amplifier Type:	Class D amplifier, SMPS
Power Output (RMS):	800 W ((200 W + 200 W) × 2)
Input Sensitivity:	- 4 dBV
Signal Processing:	DSP, 48 kHz, 24 bits
Protection:	Multiple limiter, short circuit, overheating
Controls:	On / Off switch, Main level, Full / SAT switch, Feedback destroy switch, Feedback destroy knob, High shelf knob
Indicators:	On, signal, limit, protect
Input / Output:	Input Neutrik XLR / Router-output Neutrik XLR, speaker out Neutrik NL- 4
Dimensions (W × H × D) :	483 × 66 × 347 mm
Weight:	6.5 kg
M-F3 AMP only can be used for the M-F3 SAT.	

MANUFACTURER 'S DECLARATIONS:

LIMITED WARRANTY

This limited warranty applies to the **Sennheiser** **AUDIOTECHNIK** branded products.

The statutory warranty rights towards the seller are not affected by this guarantee. In fact, it justifies additional independent warranty claims towards **Sennheiser**.

Sennheiser warrants that the **Sennheiser** product you have purchased from **Sennheiser** or from an **Sennheiser** authorized reseller is free from defects in materials or workmanship under normal use for a period of 1 year from the date of purchase.

The limited warranty period starts on the date of purchase. In order to receive warranty services, you are required to provide proof of the purchase date. Your dated sales or delivery receipt, showing the date of purchase, is your proof of the purchase date. Should products of the brands named above be in need of repair within the limited warranty period, you are entitled to warranty services according to the terms and conditions stated in this document.

This limited warranty extends only to the original purchaser of this **Sennheiser** branded product and is not transferable to anyone who obtains ownership of the **Sennheiser** branded product from the original purchaser. During the limited warranty period, **Sennheiser** will repair or replace the defective component parts or the product. All component parts or hardware products removed under this limited warranty become the property of **Sennheiser**.

In the unlikely event that your **Sennheiser** product has a recurring failure, **Sennheiser**, at its discretion, may elect to provide you with a replacement unit of **Sennheiser**'s choice that is at least equivalent to your **Sennheiser** branded product in hardware performance.

Sennheiser does not warrant that the operation of this product will be uninterrupted or error-free. **Sennheiser** is not responsible for damage that occurs as a result of your failure to follow the instructions included with the **Sennheiser** branded product.

This limited warranty does not apply,

- to wear parts (e.g. accumulator)
- to any product from which the serial number has been removed or that has been damaged or rendered defective as the result of an accident
- in case of, misuse, abuse, or other external causes
- by operation outside the usage parameters stated in the user's documentation shipped with the product
- by use of spare parts not manufactured or sold by **Sennheiser**
- by modification or service by anyone other than **Sennheiser**

These terms and conditions constitute the complete and exclusive warranty agreement between you and **Sennheiser** regarding the **Sennheiser** branded product you have purchased.

MANUFACTURER 'S DECLARATIONS:

LIMITATION OF LIABILITY

If your **Sennheiser** branded hardware product fails to work as warranted above, your sole and exclusive remedy shall be repair or replacement. **Sennheiser**'s maximum liability under this limited warranty is explicitly limited to the lesser of the price you have paid for the product or the cost of repair or replacement of any hardware components that malfunction in conditions of normal use.

Sennheiser is not liable for any damages caused by the product or the failure of the product, including any lost profits or savings or special, incidental or consequential damages. **Sennheiser** is not liable for any claim made by a third party or made by you for a third party.

This limitation of liability applies whether damages are sought, or claims are made, under this limited warranty or as a tort claim (including negligence and strict product liability), a contract claim, or any other claim. This limitation of liability cannot be waived or amended by any person. This limitation of liability will be effective even if you have advised **Sennheiser** of an authorized representative of **Sennheiser** of the possibility of any such damages. This limitation of liability however, will not apply to claims for personal injury.

This limited warranty gives you specific legal rights. You may also have other rights that may vary from state to state or from country to country. You are advised to consult applicable state or country laws for a full determination of your rights.

REQUESTING WARRANTY-SERVICE

To request warranty service for the product, contact **Sennheiser** or the **Sennheiser** authorized reseller from which you purchased the product.

EC DECLARATION OF CONFORMITY

These devices meet the essential requirements and further relevant specifications of Directives 2004/108/EC (EMC) and 2006/95/EC. For more information, see www.se-audiotechnik.com.

CORRECT DISPOSAL OF THIS PRODUCT (ELECTRICAL WASTE)

(Applicable in the European Union and other European countries with separate collection systems)



This marking, shown on the product or its literature, indicates that it should not be disposed with other household wastes at the end of its working life. To prevent possible harm to the environment or human health from uncontrolled waste disposal, please separate this from other types of wastes and recycle it responsibly to promote the sustainable reuse of material resources.

Household users should contact either the retailer where they purchased this product, or their local government office, for details on where and how they can recycle this item in an environmentally friendly manner.

Business users should contact their supplier and check the terms and conditions of the purchase contract. This product should not be mixed with other commercial wastes for disposal

MANUFACTURER 'S DECLARATIONS:

WEEE-DECLARATION



Your s)))e-systems product was developed and manufactured with high quality materials and components which can be recycled and / or reused. This symbol indicates that electrical and electronic equipment must be disposed of separately from normal waste at the end of its operational lifetime.

Please dispose of this product by bringing it to your local collection point or recycling centre for such equipment. This will help to protect the environment in which we all live.

BATTERIES AND ACCUMULATORS



The supplied batteries or rechargeable batteries can be recycled. Please dispose of them as special waste or return them to your specialist dealer. In order to protect the environment, only dispose exhausted batteries.

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